MCU

文件: MCU.kicad_sch

POWER_SHARED

文件: power.kicad_sch

FOC_M1

文件: FOC_M1.kicad_sch

FOC_M2

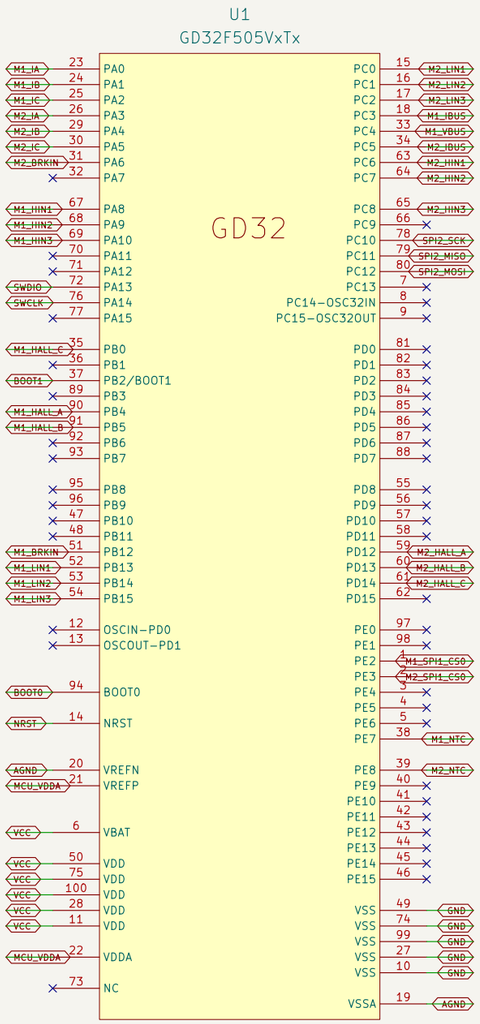
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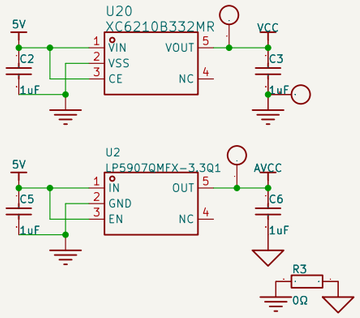
Dual independent motor framework
Common input supply + GD32F505VxTx controller
Review protection thresholds and ratings before energizing.

Sheet: /
File: GD32F5_FOC.kicad_sch

**Title:**

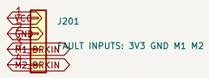
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| KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | | Id: 1/5 |

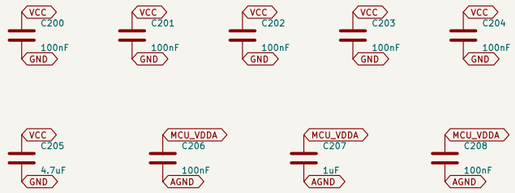






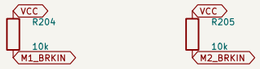






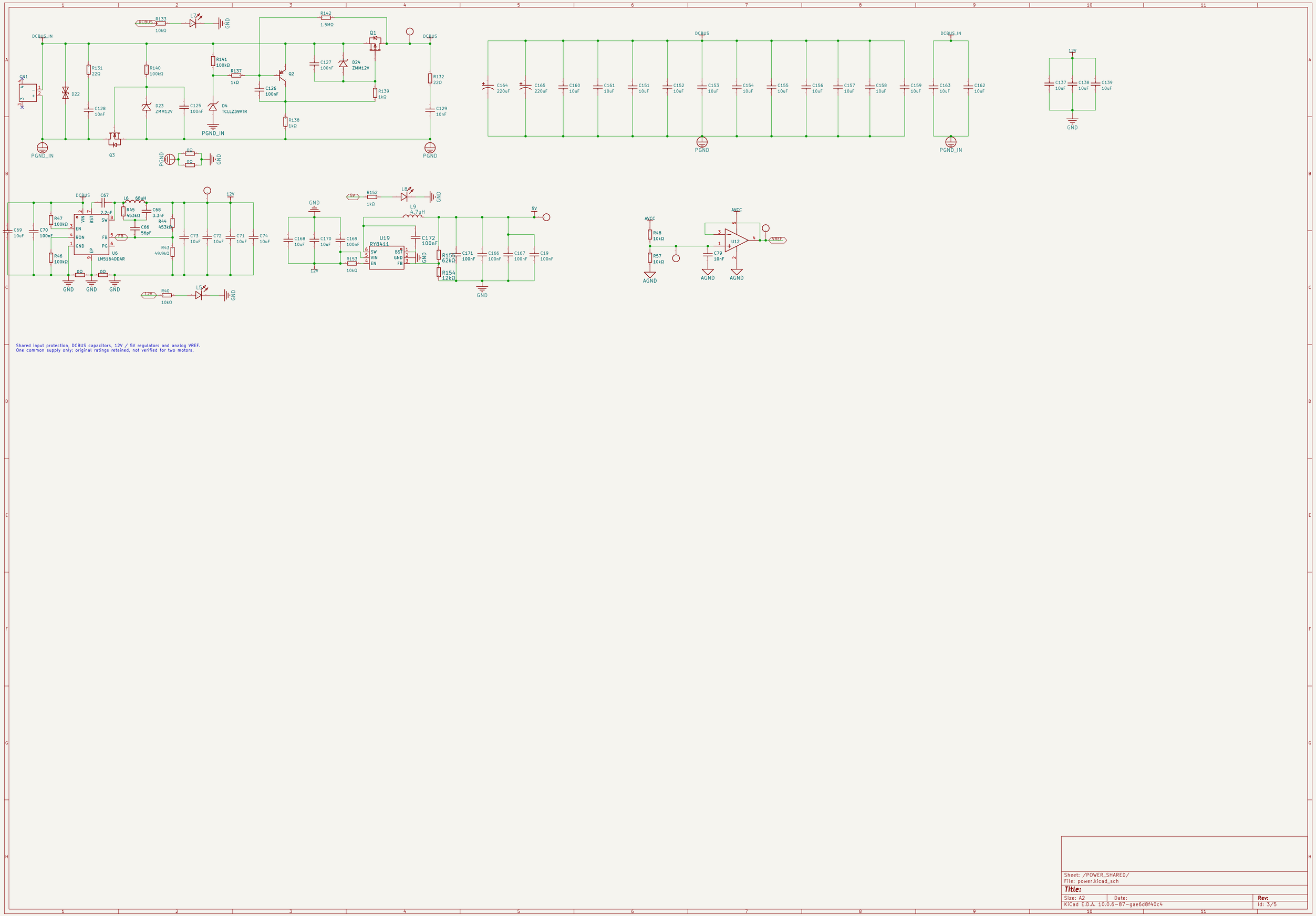



MCU_VDDA = VCC through R203 (0R), bypass C206..C208 to AGND.
AVCC LDO remains for external analog amplifiers only.
J201: 1=3V3, 2=GND, 3=M1_BRKIN, 4=M2_BRKIN.
FAULT inputs use pullups; external open-drain HEALTH holds LOW only while healthy.
Configure each timer break polarity HIGH; unplugged defaults to brake. No autonomous trip comparator fitted.
Current limits/thresholds and external fault source remain to be designed.

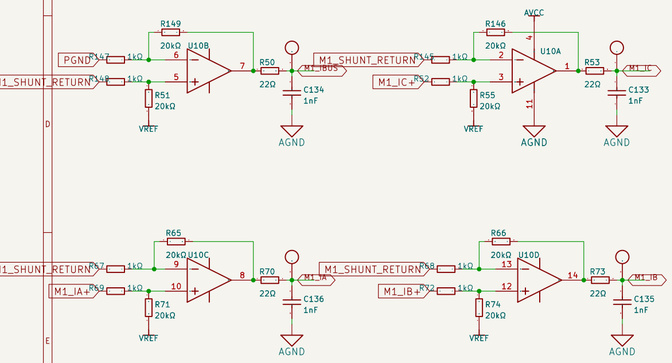
M1 TIMER0: PA8/9/10 AF1; PB13/14 AF2, PB15 AF1. BRKIN PB12 AF4.
M2 TIMER7: PC6/7 AF3, PC8 AF2; PC0/1/2 AF0. BRKIN PA6 AF4.
M1 currents PA0/1/2; M2 currents PA3/4/5 (ADC0/ADC1).
IBUS: PC3 / PC5; VBUS PC4; NTC PE7 / PE8 (ADC2).
Hall M1 PB4/5/0 TIMER2; M2 PD12/13/14 TIMER3. Release PB4 JTAG, keep SWD.
Shared SPI2 PC10/11/12 AF2; GPIO chip selects PE2 / PE3.
VA/VB/VC/COM and M2_VBUS stay on power-sheet testpoints, not MCU ADC.

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Shared input protection, DCBUS capacitors, 12V / 5V regulators and analog VREF.
One common supply only; original ratings retained, not verified for two motors.

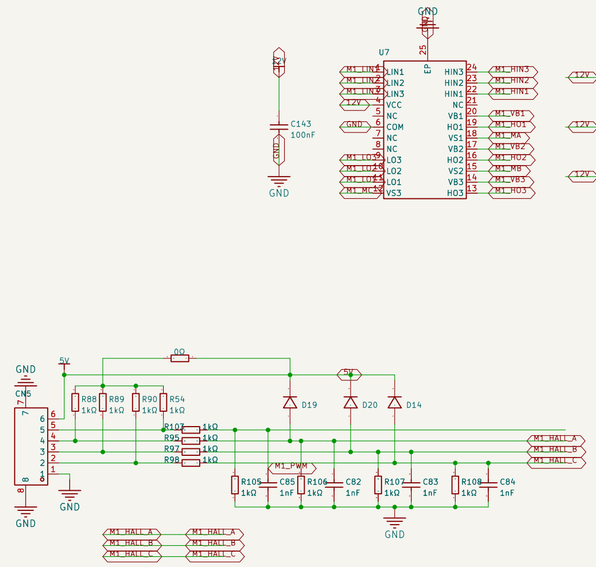
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| Sheet: /POWER_SHARED/ | |
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| KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | **Rev:** |
| | Id: 3/5 |

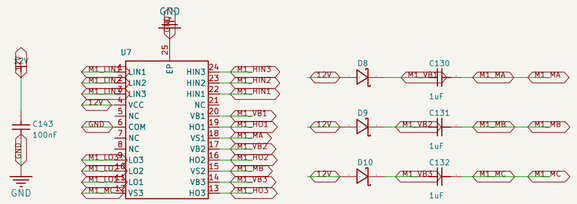






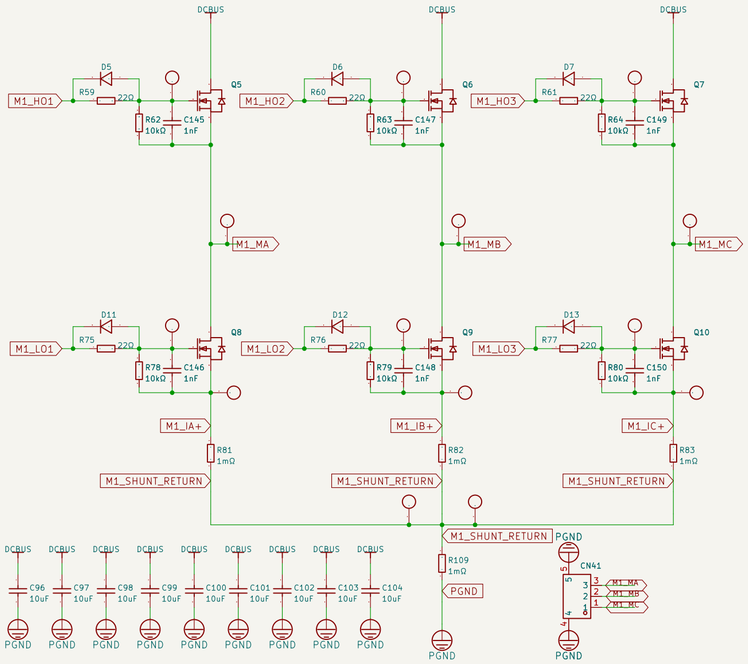


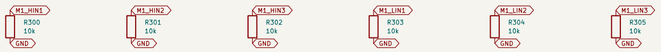









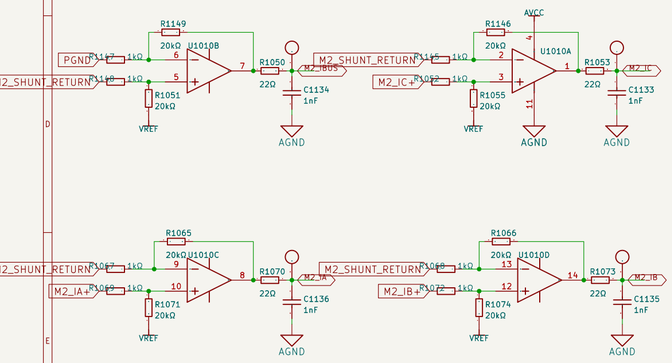

MOTOR 1: independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M1_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

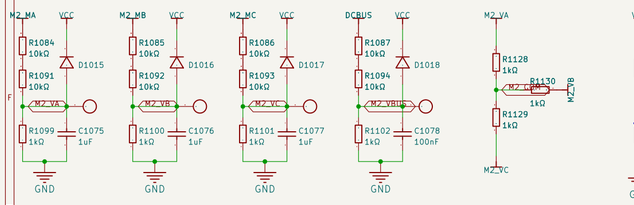
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

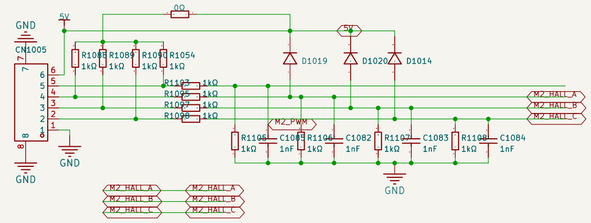
Sheet: /FOC_M1/
File: FOC_M1.kicad_sch

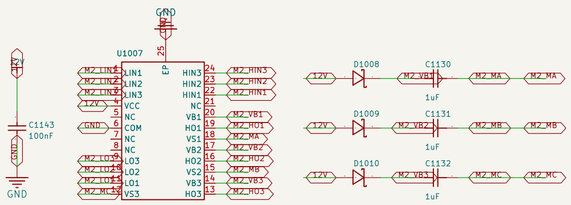
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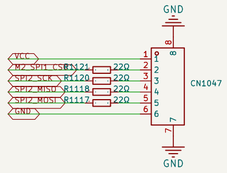
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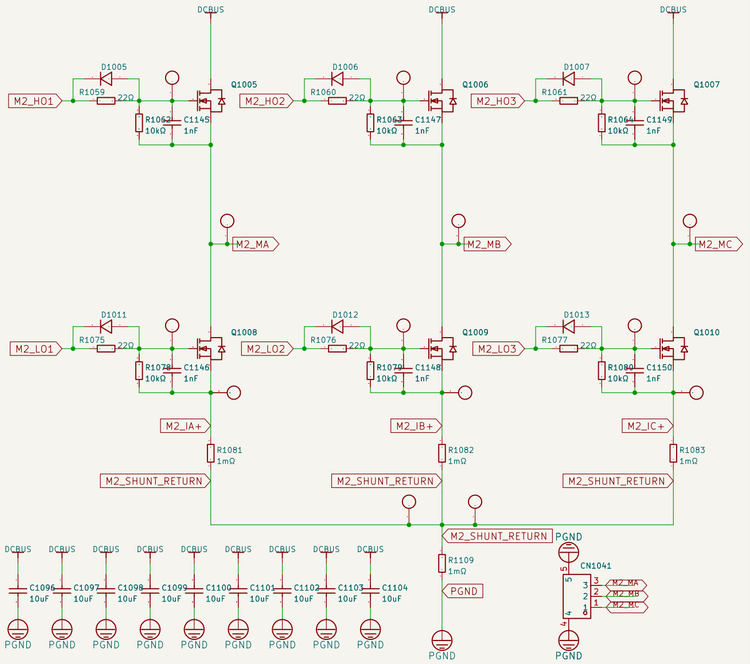


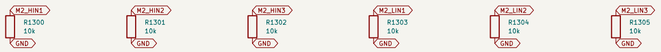




MOTOR 2: independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M2_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

Sheet: /FOC_M2/
File: FOC_M2.kicad_sch

**Title:**

| Size: A2 | Date: | Rev: |
|---|---|---|
| KiCad E.D.A. 10.0.6-87-gae6d8f40c4 | | Id: 5/5 |